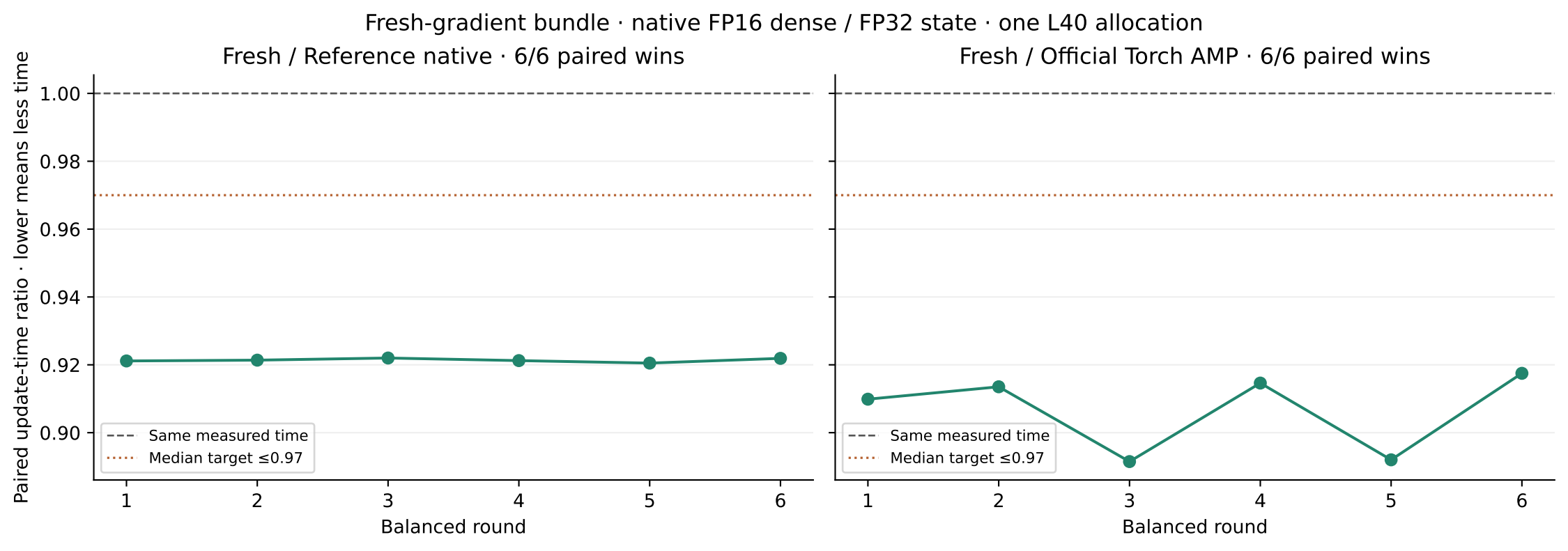
Fresh-gradient bundle · native FP16 dense / FP32 state · one L40 allocation
Fresh / Reference native · 6/6 paired wins
Fresh / Official Torch AMP · 6/6 paired wins
Paired update-time ratio · lower means less time
1.00
0.98
0.96
0.94
0.92
0.90
Same measured time
Median target ≤0.97
1
2
3
4
5
6
Balanced round